

MEET PATEL

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EDUCATION

Fairleigh Dickinson University, New Jersey

May 2026

Master's in Computer Science

Coursework: Data Mining, Artificial Intelligence, Generative AI

Ganpat University, India

Master's & Bachelor's in Computer Engineering

2019 - 2024

Coursework: Programming with Python, Artificial Intelligence, Data Science and Machine Learning

SKILLS

- **Languages:** Python, SQL, JavaScript
 - **Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, CUDA, React JS
 - **AI & Architectures:** Multi-Agent Systems (A2A), Model Context Protocol (MCP), RAG, Transformers
 - **Tools & Cloud:** Git, Docker, GCP, Kubernetes, Jenkins
- Certifications: 7+ Courses from HarvardX, Microsoft, Google, Anthropic (Focus: AI, Data Science).

PROJECTS

Privacy Preserving Yoga Recommendation Model

PyTorch, GNN, on-device computation

- Architected a **privacy centric** AI pipeline that prescribes personalized yoga routines based on real time HRV, Motion and biometric data. Utilized **100% on-device computation** to ensure data isolation.
- Engineered a **Knowledge Graph** containing **190+ yoga poses** from **50+ unstructured textbooks (PDFs)**, resolving semantic conflicts.
- Optimized a GNN for mobile deployment by implementing a **knowledge distillation pipeline**, generating a lightweight **Core ML** model optimized for the **Apple Neural Engine (ANE)**.

Real-Time Multimodal Q&A System

PyTorch, VLM, ONNX Runtime

- Built a live multimodal question answering system using NLP achieving **<100ms TTFT** by optimizing a **LLaVA style pipeline** with Qwen2-0.5B-Instruct and **Apple's FastVLM**.
- Optimized the model using **INT8 quantization**, **NPU acceleration** (CoreML) and a Python-based **ONNX pipeline**, achieving a total **4x latency reduction**.

Neuromorphic vs. Transformer NLP Benchmarking

PyTorch, snnTorch, Transformers (Hugging Face)

- Conducted accurate performance benchmarking for **NLP** while comparing a pre-trained **DistilBERT** and a custom-built, from-scratch Spiking Neural Network (SNN) on a **20,000-sample** text classification task.
- Fine-Tuned SNN for balancing the accuracy and computation power trade-off, the SNN achieved **81% of BERT's accuracy** while being **48x more computationally efficient**.

EXPERIENCE

Research Assistant

January 2025 to Present

Fairleigh Dickinson University, New Jersey

- Led research to **resolve complex spatial dependencies** (pixel to pixel relationships) in computer vision models. Performed research on Graph Neural Networks (GNN) and its hybrid versions with CNN, Transformers and Vision Transformer (ViT). Resulting in a paper acceptance at **IEEE ISEC'26 conference**.
- Processed **10,000+ patient records** and benchmarked **3 SOTA models** (EfficientNetV2, Vision Transformers and VGG16), developed a high precision pipeline using VGG16, improved accuracy **from 81% to 97.75%** results submitted to **Elsevier Pattern Recognition**.

Machine Learning Intern

January 2023 to April 2023

Tri State Technologies, Ahmedabad, India

- Streamlined **data preprocessing pipelines**, improved training data quality, and reduced model noise.
- Supported the team by developing **ML APIs using FastAPI** and **Jenkins** for CI/CD to connect model inference with frontend interface, ensuring **scalable deployment**.

Frontend Developer

September 2024 to December 2024

Vikartr Technologies, Gandhinagar, India

- Built websites using tech stack of React JS, Tailwind CSS and Node.js. Practiced core software engineering and UI/UX principles.
- Collaborated with cross functional teams of data and backend engineers.